



Instructor Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 15 Minutes

Instructor Solution Quiz No 5

[20 points]

Note - Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [20 points] Joint Distribution with two Discrete Random Variables

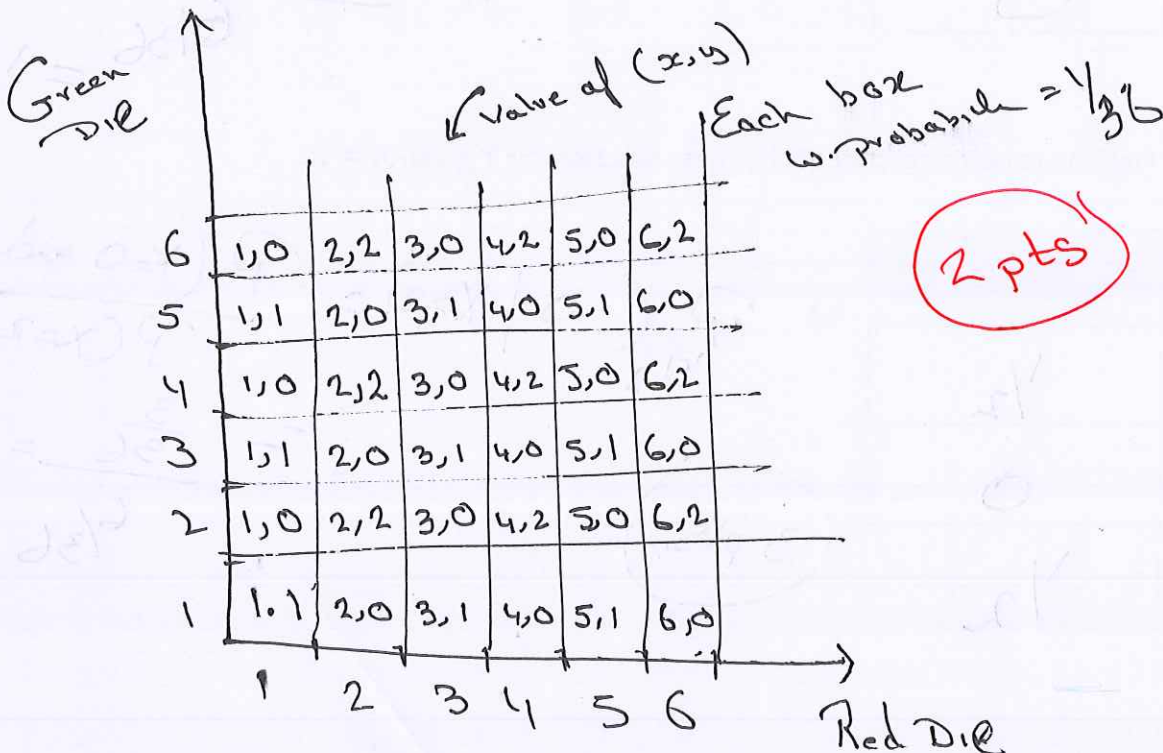
1. Suppose you roll two fair six-sided red die and a green die. Both dice have numbers one to six. We define the following two random variables X & Y:

X = number shown on the fair six-sided red die.

$$Y = \begin{cases} 0 & \text{if the two dice have one odd number and one even number.} \\ 1 & \text{if the two dice both have odd numbers.} \\ 2 & \text{if the two dice both have even numbers.} \end{cases}$$

(a) [8 points] Complete the following table for the joint probability mass function for X & Y.

Write $P_{X,Y}(x, y)$		X =						
		1	2	3	4	5	6	
Y =	0	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	→ 2pts
	1	$\frac{3}{36}$	0	$\frac{3}{36}$	0	$\frac{3}{36}$	0	→ 2pts
	2	0	$\frac{3}{36}$	0	$\frac{3}{36}$	0	$\frac{3}{36}$	→ 2pts



(b) [3 points] Complete the following table for the marginal probability mass function for RV Y.

Y	$P_Y(y)$
0	$\frac{18}{36} = \frac{1}{2}$
1	$\frac{9}{36} = \frac{1}{4}$
2	$\frac{9}{36} = \frac{1}{4}$

1 pt

1 pt

1 pt

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$P_Y(0) = \frac{18}{36}$$

$$P_Y(1) = \frac{9}{36}$$

$$P_Y(2) = \frac{9}{36}$$

(c) [3 points] Compute and write the expected value, $E[Y]$.

$$E[Y] = 0 \times \frac{1}{2} + 1 \times \frac{1}{4} + 2 \times \frac{1}{4} = \frac{1}{4} + \frac{1}{2} = 0.75$$

3 pts

(d) [3 points] Find the conditional probability mass function for RV Y given $X = 1$.

Y	$P_{Y X}(y X = 1)$
0	$\frac{1}{2}$
1	$\frac{1}{2}$
2	0

2 pt

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

1 pt

$$= \frac{P(Y=y \text{ and } X=1)}{P(X=1) = \frac{6}{36}}$$

$$\Rightarrow \frac{P(Y=0 \text{ and } X=1) = \frac{3}{36}}{\frac{6}{36}} = \frac{1}{2}$$

(e) [3 points] Find the conditional probability mass function for Y given $X = 2$.

Y	$P_{Y X}(y X = 2)$
0	$\frac{1}{2}$
1	0
2	$\frac{1}{2}$

$$P_{Y|X}(0 | X=2) = \frac{P(Y=0 \text{ and } X=2)}{P(X=2)}$$

$$= \frac{\frac{3}{36}}{\frac{6}{36}} = \frac{1}{2}$$

3 pts